

project_notebook

March 30, 2025

Project code

```
[1]: import json
import math
import pandas as pd
import numpy as np
import re
import csv
import nltk
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.svm import SVC, LinearSVC
import spacy
import pathlib

from nltk.sentiment import vader
from nltk.sentiment.vader import SentimentIntensityAnalyzer
from sklearn.datasets import load_files
from nltk.corpus import stopwords
from nltk.tokenize import word_tokenize
from wordcloud import WordCloud
from sklearn.metrics import accuracy_score
from sklearn.metrics import classification_report, confusion_matrix
from simpletransformers.classification import ClassificationModel, \
    ClassificationArgs
from sklearn.feature_extraction.text import CountVectorizer, TfidfVectorizer
from sklearn.feature_extraction import DictVectorizer
from sklearn.naive_bayes import MultinomialNB
from collections import Counter
```

0.1 Preprocessing for NERC dataset

```
[46]: cwd = pathlib.Path.cwd()
nerc_folder = cwd.joinpath('nerc_conll2003')
print('path:', nerc_folder)

# Dataset link: https://www.kaggle.com/datasets/juliangarratt/conll2003-dataset
```

path: /Users/robertostoica/Desktop/University_files_(year_3)_extra/period_4/Text

```
[47]: file_name = ('eng.train')
file_path = f"{nerc_folder}/{file_name}"

training_features = []
training_gold_labels = []

with open(file_path, "r", encoding="utf-8") as f:
    for line in f:
        # Skip empty lines (sentence separators)
        if line.strip():
            parts = line.strip().split()
            if len(parts) == 4: # Ensuring the correct number of columns
                word, pos, chunk, ne_label = parts

                # Skip "-DOCSTART-" entries
                if word != "-DOCSTART-":
                    feature_dict = {
                        "words": word,
                        "Is_upper": word.isupper(),
                        "Is_lower": word.islower(),
                        "Is_title": word.istitle(),
                        "Is_number": word.isdigit(),
                        "starts_with_capital" : word.istitle()
                    }

# features based on slides of named entities lecture (lecture 5) on slides
↳15,16,19,20
# Dataset choice was made based on factors from the "Sequence labeling for
↳parts of speech and named entities =>
# => paper, refer to pages 7,20(evaluation of named entity recognition); also
↳from the text "2016 Maynard et al =>"
# => refer to pages 25,26,28,30(3.6 approaches to nerc), 31-32-33 (3.6.2
↳supervised learning methods for nerc) =>
# => 34-35 (3.9 performance)

training_features.append(feature_dict)
training_gold_labels.append(ne_label)

# Print first 3 examples
print(training_features[:3])
print(training_gold_labels[:3])
```

```
[{'words': 'EU', 'Is_upper': True, 'Is_lower': False, 'Is_title': False,
'Is_number': False, 'starts_with_capital': False}, {'words': 'rejects',
'Is_upper': False, 'Is_lower': True, 'Is_title': False, 'Is_number': False,
'starts_with_capital': False}, {'words': 'German', 'Is_upper': False,
```

```

'Is_lower': False, 'Is_title': True, 'Is_number': False, 'starts_with_capital':
True}]
['B-ORG', 'O', 'B-MISC']

```

```

[9]: #nerc_test_path = 'NER-test.tsv'

#csv_table = pd.read_table(nerc_test_path, sep='\t')
#csv_table.to_csv('NER-test.csv', index=False)

```

```

[48]: file_path = ('NER-test.csv')

test_features = []
test_gold_labels = []

label_mapping = {
    "B-PERSON": "B-PER",
    "I-PERSON": "I-PER",
    "B-WORK_OF_ART": "B-MISC",
    "I-WORK_OF_ART": "I-MISC"
}

with open(file_path, "r", encoding="utf-8") as f:

    reader = csv.reader(f, delimiter=',')

    for line in reader:

        if len(parts) == 4 and 'sentence_id' not in line:
            sentence_id, token_id, word, ne_label = line

            mapped_label = label_mapping.get(ne_label, ne_label)

            feature_dict = {
                "words": word,
                "Is_upper": word.isupper(),
                "Is_lower": word.islower(),
                "Is_title": word.istitle(),
                "Is_number": word.isdigit(),
                "starts_with_capital" : word.istitle()
            }

            test_features.append(feature_dict)
            test_gold_labels.append(mapped_label)

# Print first 3 examples
print(test_features[:3])
print(test_gold_labels[:3])

```

```
[{'words': 'I', 'Is_upper': True, 'Is_lower': False, 'Is_title': True,
'Is_number': False, 'starts_with_capital': True}, {'words': 'would', 'Is_upper':
False, 'Is_lower': True, 'Is_title': False, 'Is_number': False,
'starts_with_capital': False}, {'words': '\n\t', 'Is_upper': False, 'Is_lower':
True, 'Is_title': False, 'Is_number': False, 'starts_with_capital': False}]
['0', '0', '0']
```

```
[49]: print(type(training_features), type(training_gold_labels), type(test_features),
↪type(test_gold_labels))
```

```
<class 'list'> <class 'list'> <class 'list'> <class 'list'>
```

0.2 Dataset statistics for NERC dataset

```
[50]: print(f'Length of training list: {len(training_features)}\n')
print(f'Length of test list: {len(test_features)}\n')
print(f'Frequency distribution of NERC labels in training list:
↪{Counter(training_gold_labels)}\n')
print(f'Frequency distribution of NERC labels in test list a:
↪{Counter(test_gold_labels)}\n')

training_label_counts = Counter(training_gold_labels)
test_label_counts = Counter(test_gold_labels)

def calculate_ratios(label_counts, total_count):
    return {label: count / total_count for label, count in label_counts.items()}

training_label_ratios = calculate_ratios(training_label_counts,
↪len(training_gold_labels))
test_label_ratios = calculate_ratios(test_label_counts, len(test_gold_labels))

print('-' * 108)

print(f'Label ratios in training list: {training_label_ratios}\n')

print(f'Label ratios in test list: {test_label_ratios}\n')
```

Length of training list: 203621

Length of test list: 237

Frequency distribution of NERC labels in training list: Counter({'0': 169578, 'B-LOC': 7140, 'B-PER': 6600, 'B-ORG': 6321, 'I-PER': 4528, 'I-ORG': 3704, 'B-MISC': 3438, 'I-LOC': 1157, 'I-MISC': 1155})

Frequency distribution of NERC labels in test list a: Counter({'0': 186, 'B-PER': 11, 'I-MISC': 10, 'B-MISC': 9, 'I-PER': 8, 'B-LOC': 7, 'B-ORG': 3, 'I-ORG': 2, 'I-LOC': 1})

Label ratios in training list: {'B-ORG': 0.03104296708099852, 'O': 0.832811939829389, 'B-MISC': 0.016884309575142052, 'B-PER': 0.0324131597428556, 'I-PER': 0.022237392017522752, 'B-LOC': 0.03506514553999833, 'I-ORG': 0.018190658134475325, 'I-MISC': 0.00567230295499973, 'I-LOC': 0.005682125124618777}

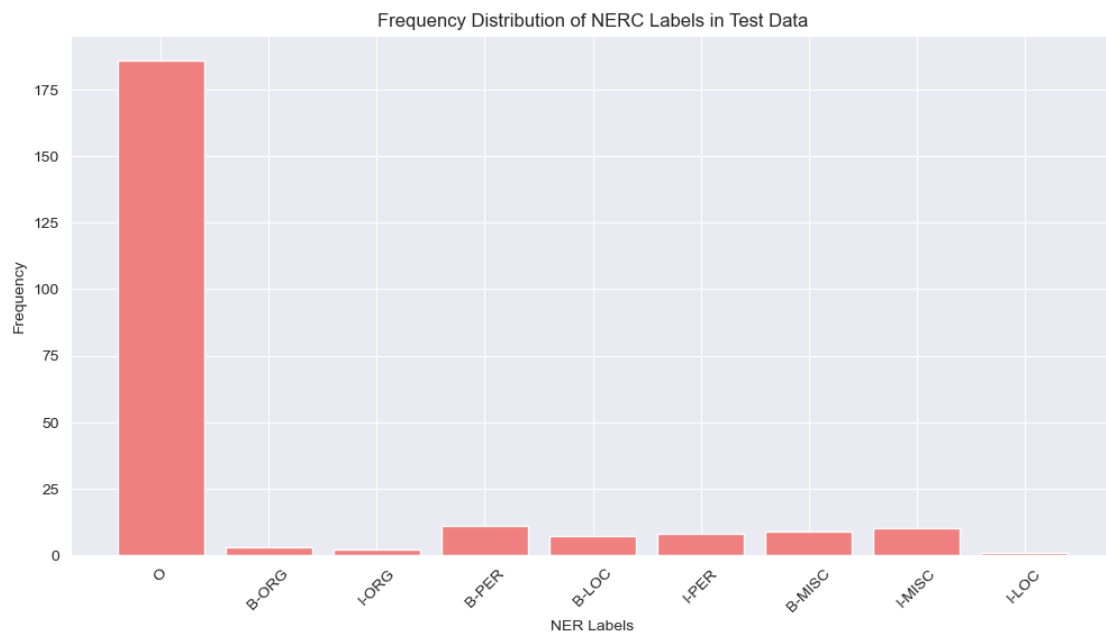
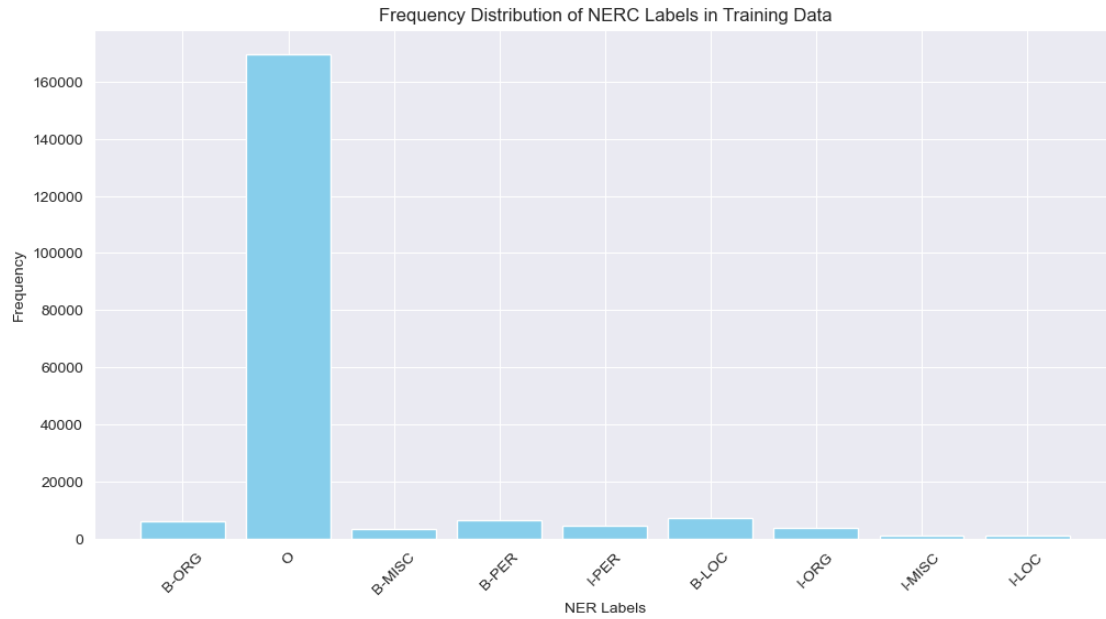
Label ratios in test list: {'O': 0.7848101265822784, 'B-ORG': 0.012658227848101266, 'I-ORG': 0.008438818565400843, 'B-PER': 0.046413502109704644, 'B-LOC': 0.029535864978902954, 'I-PER': 0.03375527426160337, 'B-MISC': 0.0379746835443038, 'I-MISC': 0.04219409282700422, 'I-LOC': 0.004219409282700422}

```
[51]: # Compute ratios for pie charts
total_training = sum(training_label_counts.values())
training_label_ratios = {label: count / total_training for label, count in
    ↪training_label_counts.items()}

total_test = sum(test_label_counts.values())
test_label_ratios = {label: count / total_test for label, count in
    ↪test_label_counts.items()}

# Bar chart for training data frequency distribution
plt.figure(figsize=(12, 6))
plt.bar(training_label_counts.keys(), training_label_counts.values(),
    ↪color='skyblue')
plt.xlabel("NER Labels")
plt.ylabel("Frequency")
plt.title("Frequency Distribution of NERC Labels in Training Data")
plt.xticks(rotation=45)
plt.show()

# Bar chart for test data frequency distribution
plt.figure(figsize=(12, 6))
plt.bar(test_label_counts.keys(), test_label_counts.values(),
    ↪color='lightcoral')
plt.xlabel("NER Labels")
plt.ylabel("Frequency")
plt.title("Frequency Distribution of NERC Labels in Test Data")
plt.xticks(rotation=45)
plt.show()
```

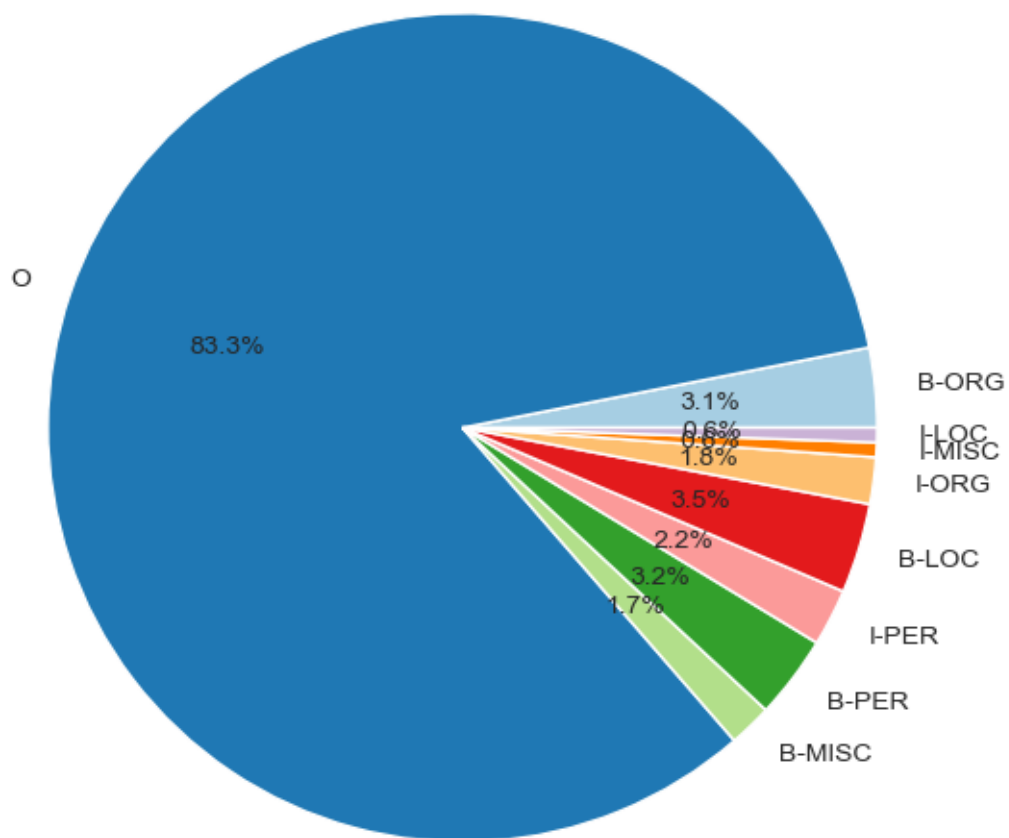


```
[52]: # Pie chart for training data percentage distribution
plt.figure(figsize=(7, 7))
plt.pie(training_label_ratios.values(), labels=training_label_ratios.keys(),
        autopct='%1.1f%%', colors=plt.cm.Paired.colors)
plt.title("Percentage Distribution of NERC Labels in Training Data")
```

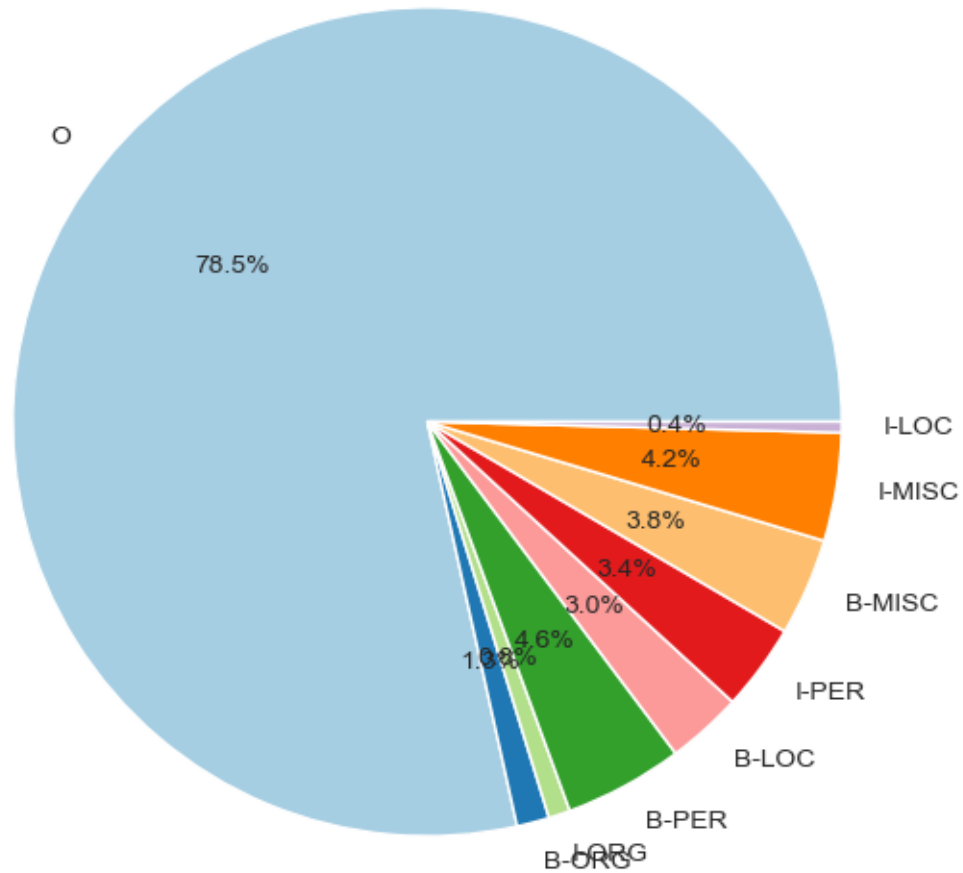
```
plt.show()

# Pie chart for test data percentage distribution
plt.figure(figsize=(7, 7))
plt.pie(test_label_ratios.values(), labels=test_label_ratios.keys(),
        autopct='%1.1f%%', colors=plt.cm.Paired.colors)
plt.title("Percentage Distribution of NERC Labels in Test Data")
plt.show()
```

Percentage Distribution of NERC Labels in Training Data



Percentage Distribution of NERC Labels in Test Data



```
[53]: print(training_features[0:3])
```

```
[{'words': 'EU', 'Is_upper': True, 'Is_lower': False, 'Is_title': False,
'Is_number': False, 'starts_with_capital': False}, {'words': 'rejects',
'Is_upper': False, 'Is_lower': True, 'Is_title': False, 'Is_number': False,
'starts_with_capital': False}, {'words': 'German', 'Is_upper': False,
'Is_lower': False, 'Is_title': True, 'Is_number': False, 'starts_with_capital':
True}]
```

```
[54]: for elem in training_features:
      print(''.join(elem['words']), elem['Is_upper'], elem['Is_lower'])
      break
```

```
EU True False
```


0.3 Naive Bayes Model for NERC

```
[55]: vec = DictVectorizer(sparse=False)

# Concatenate training and test features
all_features = training_features + test_features

# Apply vectorization
all_feature_matrix = vec.fit_transform(all_features)

# Split the vectorized features back into training and testing sets based on
↳ the number of training instances
num_train = len(training_features)
train_feature_matrix = all_feature_matrix[:num_train]
test_feature_matrix = all_feature_matrix[num_train:]

training_gold_labels = np.array(training_gold_labels)
test_gold_labels = np.array(test_gold_labels)

print(f"Train matrix: {train_feature_matrix}")
print(f"Test matrix: {test_feature_matrix}")
print()
print(f"Shape of training feature matrix: {train_feature_matrix.shape},
↳ {type(train_feature_matrix)}")
print(f"Shape of test feature matrix: {test_feature_matrix.shape},
↳ {type(test_feature_matrix)}")
print(f'Shape of training labels: {training_gold_labels.shape}')
print(f'Shape of testing labels: {test_gold_labels.shape}')
```

```
Train matrix: [[0. 0. 0. ... 0. 0. 0.]
 [1. 0. 0. ... 0. 0. 0.]
 [0. 0. 1. ... 0. 0. 0.]
```

...

```
[0. 1. 0. ... 0. 0. 0.]
 [0. 0. 1. ... 0. 0. 0.]
 [0. 1. 0. ... 0. 0. 0.]]
```

```
Test matrix: [[0. 0. 1. ... 0. 0. 0.]
 [1. 0. 0. ... 0. 0. 0.]
 [1. 0. 0. ... 0. 0. 0.]
```

...

```
[1. 0. 0. ... 0. 0. 0.]
 [1. 0. 0. ... 0. 0. 0.]
 [0. 0. 0. ... 0. 0. 0.]]
```

```
Shape of training feature matrix: (203621, 23667), <class 'numpy.ndarray'>
Shape of test feature matrix: (237, 23667), <class 'numpy.ndarray'>
Shape of training labels: (203621,)
Shape of testing labels: (237,)
```

```
[56]: nb_clf = MultinomialNB()

nb_clf.fit(train_feature_matrix, training_gold_labels)
```

```
[56]: MultinomialNB()
```

```
[57]: test_predictions = nb_clf.predict(test_feature_matrix)

accuracy = accuracy_score(test_gold_labels, test_predictions)
```

```
[58]: print(f"Test Accuracy: {accuracy:.4f}")

print("Classification Report:")
print(classification_report(test_gold_labels, test_predictions,
    ↳zero_division=0))
print("Confusion Matrix:")
print(confusion_matrix(test_gold_labels, test_predictions))
```

Test Accuracy: 0.8354

Classification Report:

	precision	recall	f1-score	support
B-LOC	0.67	0.57	0.62	7
B-MISC	0.00	0.00	0.00	9
B-ORG	0.29	0.67	0.40	3
B-PER	0.30	0.91	0.45	11
I-LOC	0.00	0.00	0.00	1
I-MISC	0.00	0.00	0.00	10
I-ORG	0.00	0.00	0.00	2
I-PER	0.00	0.00	0.00	8
0	0.96	0.98	0.97	186
accuracy			0.84	237
macro avg	0.25	0.35	0.27	237
weighted avg	0.79	0.84	0.80	237

Confusion Matrix:

```
[[ 4  0  2  1  0  0  0  0  0]
 [ 0  0  1  4  0  0  0  0  4]
 [ 0  0  2  1  0  0  0  0  0]
 [ 0  0  1 10  0  0  0  0  0]
 [ 0  0  0  1  0  0  0  0  0]
 [ 0  0  0  7  0  0  0  0  3]
 [ 2  0  0  0  0  0  0  0  0]
 [ 0  0  0  7  0  0  0  0  1]
 [ 0  1  1  2  0  0  0  0 182]]
```

0.4 Preprocessing for sentiment classification dataset

```
[59]: files_sentiment_path = ['sentiment_analysis_data/training_data_1.json',
                             'sentiment_analysis_data/training_data_2.json']

def mergeJSONfiles(filename):
    training_data_sentiment = {}
    counter = 1
    for file in filename:
        with open(file, 'r', encoding='utf-8') as sentiment_data:
            data = json.load(sentiment_data)
            for key in data:
                training_data_sentiment[str(counter)] = data[key]
                counter += 1

    with open('sentiment_analysis_data/merged_training_data.json', 'w',
              encoding='utf-8') as output_file:
        json.dump(training_data_sentiment, output_file, indent=4)

mergeJSONfiles(files_sentiment_path)
```

```
[2]: sentiment_file_path = 'sentiment_analysis_data/merged_training_data.json'

with open(sentiment_file_path, 'r') as sentiment_data:
    training_data_sentiment = json.load(sentiment_data)

print(training_data_sentiment['4000000'])
```

```
{'sentiment_label': 'positive', 'text_of_tweet': "Makes My Blood Run Red-White-And-Blue: I agree that every American should read this book -- and everybody else for that matter. I don't agree that it's scholarly. Rather, it's a joy to read -- easy to understand even for a person with two master's degrees! Between McElroy's chapter on How American Culture was Formed and Ken Burns' Lewis & Clark, I don't know which makes my blood run red-white-and-bluer. And as a child of the anti-establishment `60s, it's done a lot toward helping me understand why we Americans do what we do. It's the best history book I've ever read, the best history course I've ever taken or taught. I'm buying it for my home library for my grandchildren to use as a resource. We're also using it as a resource for a book on urban planning."}
```

```
[62]: tsv_file = 'sentiment-topic-test.tsv'
test_data_sentiment = pd.read_table(tsv_file, sep='\t')
test_data_sentiment.to_csv('testing_data_sentiment_topic.csv', index=False)
```

```
[3]: test_data_sentiment_csv = pd.read_csv('testing_data_sentiment_topic.csv')
test_data_dict = {}
test_data_dict.update(test_data_sentiment_csv['sentiment'])
print(test_data_dict)
```

```
{0: 'positive', 1: 'positive', 2: 'positive', 3: 'neutral', 4: 'neutral', 5:
'positive', 6: 'positive', 7: 'negative', 8: 'neutral', 9: 'neutral', 10:
'negative', 11: 'positive', 12: 'negative', 13: 'neutral', 14: 'negative', 15:
'neutral', 16: 'negative', 17: 'negative'}
```

```
[3]: all_text = json.load(open('sentiment_analysis_data/merged_training_data.json'))

for id_, tweet in all_text.items():
    clean_text = "".join(char if char != "\\\" else "\"" for char in_
    ↪tweet['text_of_tweet'])
    tweet['text_of_tweet'] = clean_text
    print(id_, tweet['text_of_tweet'])
    break
```

1 Great CD: My lovely Pat has one of the GREAT voices of her generation. I have listened to this CD for YEARS and I still LOVE IT. When I'm in a good mood it makes me feel better. A bad mood just evaporates like sugar in the rain. This CD just oozes LIFE. Vocals are jusat STUUNNING and lyrics just kill. One of life's hidden gems. This is a desert isle CD in my book. Why she never made it big is just beyond me. Everytime I play this, no matter black, white, young, old, male, female EVERYBODY says one thing "Who was that singing ?"

```
[ ]: def csv_to_json(csv_file_path, json_file_path):
    # Read CSV and convert to a dictionary with sentence_id as keys
    with open(csv_file_path, mode='r', encoding='utf-8') as csv_file:
        csv_reader = csv.DictReader(csv_file)
        data = {}
        for row in csv_reader:
            sentence_id = row['sentence_id']
            # Remove sentence_id from the inner dict since it's now the key
            entry = {
                "sentence": row["sentence"],
                "sentiment": row["sentiment"],
                "topic": row["topic"]
            }
            data[sentence_id] = entry

    # Write JSON file
    with open(json_file_path, mode='w', encoding='utf-8') as json_file:
        json.dump(data, json_file, indent=2)

    print(f"Successfully converted {csv_file_path} to {json_file_path}")

# File paths
csv_file = 'testing_data_sentiment_topic.csv'
json_file = 'testing_data_sentiment_topic.json'

# Convert the file
```

```
csv_to_json(csv_file, json_file)
```

0.5 Dataset statistics for sentiment classification

```
[5]: frequency_dict = {}
      freq_dict = {}

      for id_, tweet in all_text.items():
          if tweet['sentiment_label'] not in frequency_dict:
              frequency_dict[tweet['sentiment_label']] = 1
          else:
              frequency_dict[tweet['sentiment_label']] += 1

      for sentiment in test_data_dict.values():
          if sentiment not in freq_dict:
              freq_dict[sentiment] = 1
          else:
              freq_dict[sentiment] += 1

      print(f"Training data class distribution: {frequency_dict}")
      print(f"Testing data class distribution: {freq_dict}")
```

```
Training data class distribution: {'positive': 2000000, 'negative': 2000000}
Testing data class distribution: {'positive': 6, 'neutral': 6, 'negative': 6}
```

```
[6]: def entropy(labels):
      freqdist = nltk.FreqDist(labels)
      probs = [freqdist.freq(l) for l in freqdist]
      return -sum(p * math.log(p,2) for p in probs)

      print(entropy(frequency_dict))
      print(entropy(freq_dict))
```

```
1.0
1.584962500721156
```

0.6 VADER Model for sentiment classification

```
[7]: reviews_data = json.load(open('testing_data_sentiment_topic.json'))
      for id_, tweet_info in reviews_data.items():
          print(id_, tweet_info)
          break
```

```
0 {'sentence': 'The atmosphere at the stadium tonight was electric.',
'sentiment': 'positive', 'topic': 'sports'}
```

```
[11]: nltk.download('punkt')
       nltk.download('stopwords')
       nltk.download('vader_lexicon')
```

```

sia = SentimentIntensityAnalyzer()

new_words = {
    'electric': 2.0,
    'intense': 1.5,
    'hooked': 1.7,
    'whiny': -2.0,
    'poetic': 1.5,
    'mess up': -2.0,
    'wobbly': -1.0
}
sia.lexicon.update(new_words)

def preprocess_text(text):
    """Preprocess text for better sentiment analysis"""
    text = text.lower()
    text = re.sub(r'[^a-zA-Z\s]', '', text)
    tokens = word_tokenize(text)
    stop_words = set(stopwords.words('english'))
    tokens = [word for word in tokens if word not in stop_words]
    return ' '.join(tokens)

def run_vader(review):
    """
    Runs VADER sentiment analysis on a given review.
    """
    processed_review = preprocess_text(review)
    return sia.polarity_scores(processed_review)

def vader_output_to_label(vader_output):
    """
    Map VADER output to sentiment labels with optimized thresholds.
    """
    compound = vader_output['compound']

    if compound <= -0.05: # More sensitive to negative sentiment
        return 'negative'
    elif compound >= 0.05: # More sensitive to positive sentiment
        return 'positive'
    else:
        return 'neutral'

reviews = []
all_vader_output = []
gold = []

```

```

for id_, review_info in reviews_data.items():
    the_review = review_info['sentence']
    vader_output = run_vader(the_review)
    vader_label = vader_output_to_label(vader_output)

    reviews.append(the_review)
    all_vader_output.append(vader_label)
    gold.append(review_info['sentiment'])

print(classification_report(gold, all_vader_output))

print("\nMisclassified Examples:")
for i, (true_label, predicted_label) in enumerate(zip(gold, all_vader_output)):
    if true_label != predicted_label:
        print(f"\nReview: {reviews[i]}")
        print(f"True Sentiment: {true_label} | Predicted: {predicted_label}")
        scores = run_vader(reviews[i])
        print(f"VADER Scores: {scores}")
        print(f"Topic: {reviews_data[str(i)]['topic']}")

# Lecture-4-sentiment: slide 7 "Product reviews" is good for understanding the
    ↳task; slide 13 "Terminology" is good
# for understanding the language used to describe these systems; slide 21
    ↳"Sentiment Analysis Steps" gives a good
# overview of the pipeline for SA; slide 23 "Features for sentiment mining" is
    ↳good to use as a reference to choices
# which were made; slide 24 "Representing data" good to use for discussion of
    ↳why we chose this dataset due to it
# having the BIO format; slide 30 "Context Dependence" gives a good reference
    ↳for why the performance of this system
# is not as high as other systems; slide 36 "VADER lexicon" shows that VADER
    ↳uses hard coded values, so it can be
# easily modified, but it takes a lot of work

# 2016-Maynard-et-al: page 75 "Issues in opinion mining", page 76 "Polarity
    ↳recognition", page 81-83 "Methods for
# opinion mining"

```

	precision	recall	f1-score	support
negative	0.67	0.33	0.44	6
neutral	0.60	0.50	0.55	6
positive	0.60	1.00	0.75	6
accuracy			0.61	18
macro avg	0.62	0.61	0.58	18
weighted avg	0.62	0.61	0.58	18

Misclassified Examples:

Review: It's split into two timelines, which keeps it interesting but also a bit confusing at times.

True Sentiment: neutral | Predicted: positive

VADER Scores: {'neg': 0.164, 'neu': 0.603, 'pos': 0.233, 'compound': 0.2023}

Topic: book

Review: How do you concede three goals in ten minutes? The whole defence needs replacing.

True Sentiment: negative | Predicted: positive

VADER Scores: {'neg': 0.0, 'neu': 0.851, 'pos': 0.149, 'compound': 0.1027}

Topic: sports

Review: They rotated their squad for the cup game, which wasn't surprising given the schedule.

True Sentiment: neutral | Predicted: negative

VADER Scores: {'neg': 0.206, 'neu': 0.794, 'pos': 0.0, 'compound': -0.2057}

Topic: sports

Review: The trailer gave away most of the plot, but there were still a few surprises.

True Sentiment: neutral | Predicted: positive

VADER Scores: {'neg': 0.0, 'neu': 0.725, 'pos': 0.275, 'compound': 0.2263}

Topic: movie

Review: I don't get how was it supposed to work without any chemistry between the leads.

True Sentiment: negative | Predicted: neutral

VADER Scores: {'neg': 0.0, 'neu': 1.0, 'pos': 0.0, 'compound': 0.0}

Topic: movie

Review: I don't get the appeal at all, it's just a couple guys kicking a ball around at the end of the day.

True Sentiment: negative | Predicted: neutral

VADER Scores: {'neg': 0.0, 'neu': 1.0, 'pos': 0.0, 'compound': 0.0}

Topic: sports

Review: It's really incredibly impressive to mess up such a tested blockbuster formula.

True Sentiment: negative | Predicted: positive

VADER Scores: {'neg': 0.205, 'neu': 0.27, 'pos': 0.525, 'compound': 0.6936}

Topic: movie

[nltk_data] Downloading package punkt to

[nltk_data] /Users/robertostolica/nltk_data...


```
[nltk_data] Package punkt is already up-to-date!
[nltk_data] Downloading package stopwords to
[nltk_data] /Users/robertostoica/nltk_data...
[nltk_data] Package stopwords is already up-to-date!
[nltk_data] Downloading package vader_lexicon to
[nltk_data] /Users/robertostoica/nltk_data...
[nltk_data] Package vader_lexicon is already up-to-date!
```

0.7 SVM model for sentiment classification

```
[4]: nltk.download('stopwords')
stop_words = set(stopwords.words('english'))

def remove_stopwords(text):
    return " ".join([word for word in text.split() if word.lower() not in
    ↪stop_words])

def load_train_data(file_path, sample_size=500000):
    with open(file_path, 'r', encoding='utf-8') as file:
        data = json.load(file)

    X_train = [remove_stopwords(item['text_of_tweet']) for item in list(data.
    ↪values())[:sample_size]]
    y_train = [item['sentiment_label'] for item in list(data.values())[:
    ↪sample_size]]

    return X_train, y_train

def load_test_data(file_path):
    with open(file_path, 'r', encoding='utf-8') as file:
        data = json.load(file)

    X_test = [item['sentence'] for item in data.values()]
    y_test = [item['sentiment'] for item in data.values()]

    return X_test, y_test

train_data_path = 'sentiment_analysis_data/merged_training_data.json'
test_data_path = 'testing_data_sentiment_topic.json'

X_train, y_train = load_train_data(train_data_path)
X_test, y_test = load_test_data(test_data_path)

vectorizer_options = {'reviews_tfidf': TfidfVectorizer, 'review_count_bof':
    ↪CountVectorizer}
min_df_values = [2, 5, 10]
```

```

results = {}
for vec_name, VecClass in vectorizer_options.items():
    for min_df in min_df_values:
        vectorizer = VecClass(min_df=min_df, max_features=50000,
                                ngram_range=(1,2))
        X_train_vec = vectorizer.fit_transform(X_train)
        X_test_vec = vectorizer.transform(X_test)

        model = LinearSVC(class_weight='balanced', dual=False, verbose=1)
        model.fit(X_train_vec, y_train)
        y_pred = model.predict(X_test_vec)

        report = classification_report(y_test, y_pred, zero_division=1)
        results[(vec_name, min_df)] = report

for (vec_name, min_df), report in results.items():
    print(f"\n=== Results for {vec_name} with min_df={min_df} ===")
    print(report)

# Lecture-4-sentiment: slide 7 "Product reviews" is good for understanding the
    task; slide 13 "Terminology" is good
# for understanding the language used to describe these systems; slide 16 "What
    is subjectivity mining" is a good
# explanation of the task at hand and how this software can do it; slide 21
    "Sentiment Analysis Steps" gives a good
# overview of the pipeline for SA; slide 23 "Features for sentiment mining" is
    good to use as a reference to choices
# which were made; slide 24 "Representing data" good to use for discussion of
    why we chose this dataset due to it
# having the BIO format; slide 25 "Machine Learning methods" gives a reference
    as to why Linear SVM was chosen it
# was the most used and popular machine learning approach

# 2016-Maynard-et-al: page 81-83 "Methods for opinion mining"

```

[nltk_data] Downloading package stopwords to
[nltk_data] /Users/robertostoica/nltk_data...
[nltk_data] Package stopwords is already up-to-date!

[LibLinear]iter 1 act 3.313e+05 pre 3.170e+05 delta 4.076e+01 f 5.000e+05 |g|
2.664e+04 CG 9
cg reaches trust region boundary
iter 2 act 3.912e+04 pre 3.266e+04 delta 5.170e+01 f 1.687e+05 |g| 7.490e+03 CG
12
iter 3 act 1.579e+04 pre 1.345e+04 delta 5.931e+01 f 1.296e+05 |g| 5.218e+03 CG
14
iter 4 act 4.526e+03 pre 4.205e+03 delta 5.931e+01 f 1.138e+05 |g| 1.553e+03 CG

```

19
iter  5 act 5.267e+02 pre 5.114e+02 delta 5.931e+01 f 1.093e+05 |g| 4.267e+02 CG
25
iter  6 act 2.921e+01 pre 2.905e+01 delta 5.931e+01 f 1.088e+05 |g| 7.831e+01 CG
31
iter  7 act 7.279e-01 pre 7.280e-01 delta 5.931e+01 f 1.088e+05 |g| 7.988e+00 CG
36
[LibLinear]iter  1 act 3.313e+05 pre 3.170e+05 delta 4.076e+01 f 5.000e+05 |g|
2.664e+04 CG    9
cg reaches trust region boundary
iter  2 act 3.914e+04 pre 3.267e+04 delta 5.170e+01 f 1.687e+05 |g| 7.487e+03 CG
12
cg reaches trust region boundary
iter  3 act 1.621e+04 pre 1.388e+04 delta 6.221e+01 f 1.296e+05 |g| 2.992e+03 CG
16
iter  4 act 4.270e+03 pre 3.949e+03 delta 6.221e+01 f 1.134e+05 |g| 1.076e+03 CG
22
iter  5 act 3.617e+02 pre 3.506e+02 delta 6.221e+01 f 1.091e+05 |g| 4.279e+02 CG
23
iter  6 act 2.506e+01 pre 2.496e+01 delta 6.221e+01 f 1.088e+05 |g| 7.501e+01 CG
30
iter  7 act 7.578e-01 pre 7.580e-01 delta 6.221e+01 f 1.087e+05 |g| 7.297e+00 CG
37
[LibLinear]iter  1 act 3.313e+05 pre 3.170e+05 delta 4.076e+01 f 5.000e+05 |g|
2.664e+04 CG    9
cg reaches trust region boundary
iter  2 act 3.911e+04 pre 3.266e+04 delta 5.169e+01 f 1.687e+05 |g| 7.486e+03 CG
12
iter  3 act 1.525e+04 pre 1.289e+04 delta 5.575e+01 f 1.296e+05 |g| 6.037e+03 CG
13
iter  4 act 4.945e+03 pre 4.624e+03 delta 5.575e+01 f 1.144e+05 |g| 2.037e+03 CG
17
iter  5 act 6.719e+02 pre 6.534e+02 delta 5.575e+01 f 1.094e+05 |g| 2.948e+02 CG
30
iter  6 act 1.980e+01 pre 1.968e+01 delta 5.575e+01 f 1.088e+05 |g| 8.070e+01 CG
25
iter  7 act 1.382e+00 pre 1.382e+00 delta 5.575e+01 f 1.087e+05 |g| 8.429e+00 CG
36
[LibLinear]iter  1 act 2.973e+05 pre 2.789e+05 delta 3.670e+00 f 5.000e+05 |g|
3.192e+05 CG   10
cg reaches trust region boundary
iter  2 act 4.426e+04 pre 3.617e+04 delta 4.730e+00 f 2.027e+05 |g| 5.986e+04 CG
12
cg reaches trust region boundary
iter  3 act 2.170e+04 pre 1.816e+04 delta 6.268e+00 f 1.584e+05 |g| 2.554e+04 CG
11
cg reaches trust region boundary
iter  4 act 1.216e+04 pre 1.227e+04 delta 6.447e+00 f 1.367e+05 |g| 1.342e+04 CG

```

```

12
cg reaches trust region boundary
iter 5 act 1.127e+04 pre 1.018e+04 delta 7.519e+00 f 1.245e+05 |g| 1.366e+04 CG
18
cg reaches trust region boundary
iter 6 act 7.381e+03 pre 7.270e+03 delta 7.723e+00 f 1.133e+05 |g| 7.346e+03 CG
22
cg reaches trust region boundary
iter 7 act 6.023e+03 pre 5.712e+03 delta 8.428e+00 f 1.059e+05 |g| 5.954e+03 CG
27
cg reaches trust region boundary
iter 8 act 4.097e+03 pre 3.956e+03 delta 8.967e+00 f 9.987e+04 |g| 3.886e+03 CG
31
cg reaches trust region boundary
iter 9 act 2.339e+03 pre 2.827e+03 delta 8.967e+00 f 9.577e+04 |g| 2.608e+03 CG
37
cg reaches trust region boundary
iter 10 act 2.286e+03 pre 2.320e+03 delta 8.967e+00 f 9.343e+04 |g| 4.274e+03 CG
49
cg reaches trust region boundary
iter 11 act -4.847e+02 pre 1.068e+03 delta 3.653e+00 f 9.115e+04 |g| 1.511e+03
CG 66
cg reaches trust region boundary
iter 11 act 3.756e+02 pre 6.873e+02 delta 2.513e+00 f 9.115e+04 |g| 1.511e+03 CG
44
cg reaches trust region boundary
iter 12 act 5.362e+02 pre 5.405e+02 delta 2.513e+00 f 9.077e+04 |g| 2.872e+03 CG
39
cg reaches trust region boundary
iter 13 act 1.814e+02 pre 1.906e+02 delta 2.513e+00 f 9.023e+04 |g| 5.347e+02 CG
39
cg reaches trust region boundary
iter 14 act 1.523e+02 pre 1.611e+02 delta 2.513e+00 f 9.005e+04 |g| 4.970e+02 CG
43
cg reaches trust region boundary
iter 15 act -6.742e+02 pre 1.318e+02 delta 6.283e-01 f 8.990e+04 |g| 4.182e+02
CG 49
cg reaches trust region boundary
iter 15 act 2.672e+01 pre 6.256e+01 delta 4.005e-01 f 8.990e+04 |g| 4.182e+02 CG
24
cg reaches trust region boundary
iter 16 act 5.213e+01 pre 5.245e+01 delta 4.005e-01 f 8.987e+04 |g| 9.410e+02 CG
23
cg reaches trust region boundary
iter 17 act 1.669e+01 pre 1.691e+01 delta 4.005e-01 f 8.982e+04 |g| 1.828e+02 CG
21
cg reaches trust region boundary
iter 18 act 1.709e+01 pre 1.733e+01 delta 4.058e-01 f 8.981e+04 |g| 3.799e+02 CG

```

23
cg reaches trust region boundary
iter 19 act 1.649e+01 pre 1.651e+01 delta 4.191e-01 f 8.979e+04 |g| 3.136e+02 CG
24
cg reaches trust region boundary
iter 20 act 1.574e+01 pre 1.604e+01 delta 4.233e-01 f 8.977e+04 |g| 2.969e+02 CG
24
cg reaches trust region boundary
iter 21 act 1.660e+01 pre 1.662e+01 delta 4.588e-01 f 8.976e+04 |g| 3.501e+02 CG
25
cg reaches trust region boundary
iter 22 act 1.498e+01 pre 1.537e+01 delta 4.739e-01 f 8.974e+04 |g| 2.600e+02 CG
26
cg reaches trust region boundary
iter 23 act 1.539e+01 pre 1.544e+01 delta 4.831e-01 f 8.972e+04 |g| 3.777e+02 CG
27
cg reaches trust region boundary
iter 24 act 1.445e+01 pre 1.502e+01 delta 4.831e-01 f 8.971e+04 |g| 3.662e+02 CG
28
cg reaches trust region boundary
iter 25 act 1.408e+01 pre 1.415e+01 delta 4.855e-01 f 8.969e+04 |g| 2.691e+02 CG
29
cg reaches trust region boundary
iter 26 act -1.426e+02 pre 1.434e+01 delta 1.214e-01 f 8.968e+04 |g| 3.162e+02
CG 30
cg reaches trust region boundary
iter 26 act 9.900e-01 pre 6.621e+00 delta 6.069e-02 f 8.968e+04 |g| 3.162e+02 CG
16
cg reaches trust region boundary
iter 27 act 6.404e+00 pre 6.406e+00 delta 6.086e-02 f 8.968e+04 |g| 3.718e+02 CG
13
cg reaches trust region boundary
iter 28 act 1.217e+00 pre 1.219e+00 delta 6.455e-02 f 8.967e+04 |g| 2.329e+02 CG
10
cg reaches trust region boundary
iter 29 act 1.566e+00 pre 1.586e+00 delta 6.455e-02 f 8.967e+04 |g| 1.124e+02 CG
12
cg reaches trust region boundary
iter 30 act -9.461e-02 pre 1.736e+00 delta 3.134e-02 f 8.967e+04 |g| 1.462e+02
CG 10
cg reaches trust region boundary
iter 30 act 9.224e-01 pre 1.026e+00 delta 3.134e-02 f 8.967e+04 |g| 1.462e+02 CG
9
cg reaches trust region boundary
iter 31 act 7.484e-01 pre 7.495e-01 delta 3.200e-02 f 8.967e+04 |g| 1.343e+02 CG
9
cg reaches trust region boundary
iter 32 act 7.816e-01 pre 7.818e-01 delta 3.378e-02 f 8.967e+04 |g| 1.226e+02 CG

```

10
cg reaches trust region boundary
iter 33 act 8.245e-01 pre 8.269e-01 delta 3.724e-02 f 8.967e+04 |g| 1.292e+02 CG
10
cg reaches trust region boundary
iter 34 act 8.986e-01 pre 8.992e-01 delta 4.276e-02 f 8.967e+04 |g| 1.056e+02 CG
10
cg reaches trust region boundary
iter 35 act 9.811e-01 pre 9.850e-01 delta 4.822e-02 f 8.967e+04 |g| 1.059e+02 CG
10
cg reaches trust region boundary
iter 36 act 1.089e+00 pre 1.091e+00 delta 5.207e-02 f 8.967e+04 |g| 1.157e+02 CG
11
cg reaches trust region boundary
iter 37 act 1.223e+00 pre 1.224e+00 delta 5.667e-02 f 8.966e+04 |g| 1.562e+02 CG
11
cg reaches trust region boundary
iter 38 act 1.294e+00 pre 1.304e+00 delta 6.077e-02 f 8.966e+04 |g| 1.526e+02 CG
12
cg reaches trust region boundary
iter 39 act 1.348e+00 pre 1.339e+00 delta 6.269e-02 f 8.966e+04 |g| 1.446e+02 CG
13
cg reaches trust region boundary
iter 40 act -5.452e-01 pre 1.588e+00 delta 2.640e-02 f 8.966e+04 |g| 1.427e+02
CG 10
cg reaches trust region boundary
iter 40 act 7.537e-01 pre 8.806e-01 delta 2.640e-02 f 8.966e+04 |g| 1.427e+02 CG
8
cg reaches trust region boundary
iter 41 act 6.188e-01 pre 6.194e-01 delta 2.889e-02 f 8.966e+04 |g| 1.017e+02 CG
9
cg reaches trust region boundary
iter 42 act 6.314e-01 pre 6.315e-01 delta 3.258e-02 f 8.966e+04 |g| 8.190e+01 CG
9
cg reaches trust region boundary
iter 43 act 6.915e-01 pre 6.936e-01 delta 3.598e-02 f 8.966e+04 |g| 8.114e+01 CG
9
cg reaches trust region boundary
iter 44 act 7.971e-01 pre 7.988e-01 delta 4.094e-02 f 8.966e+04 |g| 1.161e+02 CG
10
cg reaches trust region boundary
iter 45 act 8.566e-01 pre 8.581e-01 delta 4.593e-02 f 8.966e+04 |g| 8.682e+01 CG
10
cg reaches trust region boundary
iter 46 act 9.492e-01 pre 9.554e-01 delta 4.953e-02 f 8.966e+04 |g| 9.645e+01 CG
11
cg reaches trust region boundary
iter 47 act 1.008e+00 pre 1.010e+00 delta 5.023e-02 f 8.965e+04 |g| 1.259e+02 CG

```

```

12
cg reaches trust region boundary
iter 48 act 1.093e+00 pre 1.103e+00 delta 5.135e-02 f 8.965e+04 |g| 1.047e+02 CG
12
cg reaches trust region boundary
iter 49 act 1.108e+00 pre 1.112e+00 delta 5.412e-02 f 8.965e+04 |g| 1.248e+02 CG
13
cg reaches trust region boundary
iter 50 act 1.151e+00 pre 1.162e+00 delta 5.870e-02 f 8.965e+04 |g| 9.682e+01 CG
13
cg reaches trust region boundary
iter 51 act 1.249e+00 pre 1.253e+00 delta 6.665e-02 f 8.965e+04 |g| 1.140e+02 CG
13
cg reaches trust region boundary
iter 52 act 1.333e+00 pre 1.347e+00 delta 7.426e-02 f 8.965e+04 |g| 9.040e+01 CG
13
cg reaches trust region boundary
iter 53 act 1.502e+00 pre 1.506e+00 delta 8.178e-02 f 8.965e+04 |g| 1.241e+02 CG
15
cg reaches trust region boundary
iter 54 act 1.762e+00 pre 1.779e+00 delta 9.653e-02 f 8.965e+04 |g| 3.215e+02 CG
15
cg reaches trust region boundary
iter 55 act 1.851e+00 pre 1.857e+00 delta 1.110e-01 f 8.964e+04 |g| 1.287e+02 CG
15
cg reaches trust region boundary
iter 56 act 2.192e+00 pre 2.215e+00 delta 1.275e-01 f 8.964e+04 |g| 3.716e+02 CG
16
cg reaches trust region boundary
iter 57 act 2.275e+00 pre 2.286e+00 delta 1.343e-01 f 8.964e+04 |g| 1.411e+02 CG
17
cg reaches trust region boundary
iter 58 act 2.404e+00 pre 2.426e+00 delta 1.343e-01 f 8.964e+04 |g| 1.591e+02 CG
20
cg reaches trust region boundary
iter 59 act 2.728e+00 pre 2.735e+00 delta 1.490e-01 f 8.964e+04 |g| 1.646e+02 CG
20
cg reaches trust region boundary
iter 60 act 2.281e+00 pre 2.313e+00 delta 1.493e-01 f 8.963e+04 |g| 1.168e+02 CG
20
cg reaches trust region boundary
iter 61 act 2.568e+00 pre 2.574e+00 delta 1.508e-01 f 8.963e+04 |g| 1.697e+02 CG
23
cg reaches trust region boundary
iter 62 act 2.522e+00 pre 2.552e+00 delta 1.573e-01 f 8.963e+04 |g| 1.546e+02 CG
23
cg reaches trust region boundary
iter 63 act 2.355e+00 pre 2.363e+00 delta 1.608e-01 f 8.963e+04 |g| 1.039e+02 CG

```

23
 cg reaches trust region boundary
 iter 64 act 2.324e+00 pre 2.353e+00 delta 1.608e-01 f 8.962e+04 |g| 1.615e+02 CG
 25
 cg reaches trust region boundary
 iter 65 act 2.470e+00 pre 2.483e+00 delta 1.722e-01 f 8.962e+04 |g| 1.337e+02 CG
 25
 cg reaches trust region boundary
 iter 66 act 2.189e+00 pre 2.213e+00 delta 1.763e-01 f 8.962e+04 |g| 1.034e+02 CG
 25
 cg reaches trust region boundary
 iter 67 act 2.383e+00 pre 2.403e+00 delta 1.881e-01 f 8.962e+04 |g| 1.153e+02 CG
 27
 cg reaches trust region boundary
 iter 68 act 2.261e+00 pre 2.279e+00 delta 1.992e-01 f 8.961e+04 |g| 1.219e+02 CG
 28
 cg reaches trust region boundary
 iter 69 act 2.327e+00 pre 2.352e+00 delta 2.138e-01 f 8.961e+04 |g| 1.278e+02 CG
 28
 cg reaches trust region boundary
 iter 70 act 2.159e+00 pre 2.190e+00 delta 2.142e-01 f 8.961e+04 |g| 1.178e+02 CG
 30
 cg reaches trust region boundary
 iter 71 act 2.295e+00 pre 2.317e+00 delta 2.228e-01 f 8.961e+04 |g| 1.152e+02 CG
 32
 cg reaches trust region boundary
 iter 72 act 2.074e+00 pre 2.103e+00 delta 2.316e-01 f 8.961e+04 |g| 8.069e+01 CG
 32
 cg reaches trust region boundary
 iter 73 act 1.926e+00 pre 1.961e+00 delta 2.325e-01 f 8.960e+04 |g| 9.613e+01 CG
 34
 cg reaches trust region boundary
 iter 74 act 1.846e+00 pre 1.885e+00 delta 2.327e-01 f 8.960e+04 |g| 1.047e+02 CG
 35
 cg reaches trust region boundary
 iter 75 act 1.636e+00 pre 1.688e+00 delta 2.327e-01 f 8.960e+04 |g| 2.059e+02 CG
 38
 cg reaches trust region boundary
 iter 76 act 1.493e+00 pre 1.542e+00 delta 2.327e-01 f 8.960e+04 |g| 9.152e+01 CG
 40
 cg reaches trust region boundary
 iter 77 act 1.342e+00 pre 1.398e+00 delta 2.327e-01 f 8.960e+04 |g| 7.192e+01 CG
 43
 cg reaches trust region boundary
 iter 78 act 1.164e+00 pre 1.214e+00 delta 2.327e-01 f 8.959e+04 |g| 5.489e+01 CG
 46
 cg reaches trust region boundary
 iter 79 act 1.044e+00 pre 1.085e+00 delta 2.327e-01 f 8.959e+04 |g| 1.707e+02 CG


```

48
cg reaches trust region boundary
iter 80 act 9.055e-01 pre 9.553e-01 delta 2.327e-01 f 8.959e+04 |g| 5.537e+01 CG
51
cg reaches trust region boundary
iter 81 act 7.483e-01 pre 7.984e-01 delta 2.327e-01 f 8.959e+04 |g| 4.219e+01 CG
56
cg reaches trust region boundary
iter 82 act 6.272e-01 pre 6.806e-01 delta 2.327e-01 f 8.959e+04 |g| 5.923e+01 CG
61
cg reaches trust region boundary
iter 83 act 5.028e-01 pre 5.491e-01 delta 2.327e-01 f 8.959e+04 |g| 2.465e+01 CG
67
cg reaches trust region boundary
iter 84 act 3.887e-01 pre 4.342e-01 delta 2.327e-01 f 8.959e+04 |g| 3.083e+01 CG
77
cg reaches trust region boundary
iter 85 act 2.861e-01 pre 3.190e-01 delta 2.327e-01 f 8.959e+04 |g| 2.355e+01 CG
91
[LibLinear]iter 1 act 2.973e+05 pre 2.789e+05 delta 3.670e+00 f 5.000e+05 |g|
3.192e+05 CG 10
cg reaches trust region boundary
iter 2 act 4.428e+04 pre 3.616e+04 delta 4.730e+00 f 2.027e+05 |g| 5.986e+04 CG
12
cg reaches trust region boundary
iter 3 act 2.169e+04 pre 1.816e+04 delta 6.265e+00 f 1.584e+05 |g| 2.475e+04 CG
11
cg reaches trust region boundary
iter 4 act 1.210e+04 pre 1.219e+04 delta 6.405e+00 f 1.367e+05 |g| 1.322e+04 CG
12
cg reaches trust region boundary
iter 5 act 1.124e+04 pre 1.015e+04 delta 7.429e+00 f 1.246e+05 |g| 1.201e+04 CG
18
cg reaches trust region boundary
iter 6 act 7.337e+03 pre 7.242e+03 delta 7.593e+00 f 1.134e+05 |g| 7.461e+03 CG
22
cg reaches trust region boundary
iter 7 act 6.020e+03 pre 5.694e+03 delta 8.296e+00 f 1.060e+05 |g| 6.168e+03 CG
27
cg reaches trust region boundary
iter 8 act 4.087e+03 pre 3.954e+03 delta 8.809e+00 f 1.000e+05 |g| 4.021e+03 CG
31
cg reaches trust region boundary
iter 9 act 2.366e+03 pre 2.825e+03 delta 8.809e+00 f 9.591e+04 |g| 2.670e+03 CG
36
cg reaches trust region boundary
iter 10 act 2.337e+03 pre 2.357e+03 delta 8.809e+00 f 9.354e+04 |g| 4.022e+03 CG
49

```

```

cg reaches trust region boundary
iter 11 act 1.063e+03 pre 1.088e+03 delta 8.809e+00 f 9.121e+04 |g| 1.460e+03 CG
63
cg reaches trust region boundary
iter 12 act 4.966e+02 pre 5.148e+02 delta 8.809e+00 f 9.014e+04 |g| 8.555e+02 CG
107
iter 13 act 8.999e+01 pre 9.078e+01 delta 8.809e+00 f 8.965e+04 |g| 3.041e+02 CG
222
iter 14 act 4.126e+00 pre 4.230e+00 delta 8.809e+00 f 8.956e+04 |g| 5.023e+01 CG
307
[LibLinear]iter 1 act 2.973e+05 pre 2.789e+05 delta 3.670e+00 f 5.000e+05 |g|
3.192e+05 CG 10
cg reaches trust region boundary
iter 2 act 4.423e+04 pre 3.617e+04 delta 4.729e+00 f 2.027e+05 |g| 5.986e+04 CG
12
cg reaches trust region boundary
iter 3 act 2.173e+04 pre 1.819e+04 delta 6.271e+00 f 1.584e+05 |g| 2.791e+04 CG
11
cg reaches trust region boundary
iter 4 act 1.180e+04 pre 1.188e+04 delta 6.271e+00 f 1.367e+05 |g| 1.384e+04 CG
13
cg reaches trust region boundary
iter 5 act 1.138e+04 pre 1.034e+04 delta 7.249e+00 f 1.249e+05 |g| 1.417e+04 CG
18
cg reaches trust region boundary
iter 6 act 7.325e+03 pre 7.187e+03 delta 7.503e+00 f 1.135e+05 |g| 8.053e+03 CG
22
cg reaches trust region boundary
iter 7 act 5.873e+03 pre 5.582e+03 delta 8.072e+00 f 1.062e+05 |g| 8.849e+03 CG
26
cg reaches trust region boundary
iter 8 act 4.144e+03 pre 3.992e+03 delta 8.569e+00 f 1.003e+05 |g| 4.976e+03 CG
31
cg reaches trust region boundary
iter 9 act 2.437e+03 pre 2.859e+03 delta 8.569e+00 f 9.616e+04 |g| 2.935e+03 CG
36
cg reaches trust region boundary
iter 10 act 2.321e+03 pre 2.337e+03 delta 8.569e+00 f 9.373e+04 |g| 6.368e+03 CG
45
cg reaches trust region boundary
iter 11 act 1.087e+03 pre 1.166e+03 delta 8.569e+00 f 9.140e+04 |g| 1.765e+03 CG
60
cg reaches trust region boundary
iter 12 act 5.820e+02 pre 6.256e+02 delta 8.569e+00 f 9.032e+04 |g| 9.366e+02 CG
92
iter 13 act 1.610e+02 pre 1.621e+02 delta 8.569e+00 f 8.974e+04 |g| 4.752e+02 CG
191
iter 14 act 9.150e+00 pre 9.270e+00 delta 8.569e+00 f 8.957e+04 |g| 7.711e+01 CG

```

=== Results for reviews_tfidf with min_df=2 ===

	precision	recall	f1-score	support
negative	0.50	0.83	0.62	6
neutral	1.00	0.00	0.00	6
positive	0.75	1.00	0.86	6
accuracy			0.61	18
macro avg	0.75	0.61	0.49	18
weighted avg	0.75	0.61	0.49	18

=== Results for reviews_tfidf with min_df=5 ===

	precision	recall	f1-score	support
negative	0.50	0.83	0.62	6
neutral	1.00	0.00	0.00	6
positive	0.75	1.00	0.86	6
accuracy			0.61	18
macro avg	0.75	0.61	0.49	18
weighted avg	0.75	0.61	0.49	18

=== Results for reviews_tfidf with min_df=10 ===

	precision	recall	f1-score	support
negative	0.50	0.83	0.62	6
neutral	1.00	0.00	0.00	6
positive	0.75	1.00	0.86	6
accuracy			0.61	18
macro avg	0.75	0.61	0.49	18
weighted avg	0.75	0.61	0.49	18

=== Results for review_count_bof with min_df=2 ===

	precision	recall	f1-score	support
negative	0.57	0.67	0.62	6
neutral	1.00	0.00	0.00	6
positive	0.55	1.00	0.71	6
accuracy			0.56	18
macro avg	0.71	0.56	0.44	18
weighted avg	0.71	0.56	0.44	18

=== Results for review_count_bof with min_df=5 ===

	precision	recall	f1-score	support
negative	0.57	0.67	0.62	6
neutral	1.00	0.00	0.00	6
positive	0.55	1.00	0.71	6
accuracy			0.56	18
macro avg	0.71	0.56	0.44	18
weighted avg	0.71	0.56	0.44	18

=== Results for review_count_bof with min_df=10 ===

	precision	recall	f1-score	support
negative	0.57	0.67	0.62	6
neutral	1.00	0.00	0.00	6
positive	0.55	1.00	0.71	6
accuracy			0.56	18
macro avg	0.71	0.56	0.44	18
weighted avg	0.71	0.56	0.44	18

0.8 Preprocessing for topic classification dataset

```
[2]: topic_path = 'big_data.csv'

topic_test_data = 'testing_data_sentiment_topic.csv'

training_data_topic = pd.read_csv(topic_path)
training_data_topic = training_data_topic.drop(columns=['Unnamed: 0'])

testing_data_topic = pd.read_csv(topic_test_data)
testing_data_topic = testing_data_topic.drop(columns=['sentence_id',
↳ 'sentiment'])

print(training_data_topic.shape)
print(training_data_topic.head())
print(testing_data_topic.shape)
print(testing_data_topic.head())

# Dataset link: https://www.kaggle.com/datasets/aronferencz/
↳ topic-analysis-dataset-sportbookmovie
```

(120307, 2)

```

                                review label
0  I'm only a third way in. Shipped lightening fa... book
1  "There have been so many times when I was scar... book
2  The media could not be loaded. I personally ha... book
3  I have been a fan of Britney's music since the... book
4  Whether or not you're a fan, it's a great read... book
(18, 2)

```

```

                                sentence  topic
0  The atmosphere at the stadium tonight was elec... sports
1  The game was so intense I forgot to breathe at... sports
2          It had me hooked from the first chapter.    book
3  It's more of a slow burn than a page-turner, b...    book
4  It's split into two timelines, which keeps it ...    book

```

```
[3]: print(training_data_topic['review'], training_data_topic['label'])
```

```

0      I'm only a third way in. Shipped lightening fa...
1      "There have been so many times when I was scar...
2      The media could not be loaded. I personally ha...
3      I have been a fan of Britney's music since the...
4      Whether or not you're a fan, it's a great read...

...
120302  VAL GARDENA, Italy (Reuters) - Max Rauffer be...
120303  Red Sox general manager Theo Epstein acknowled...
120304  The Miami Dolphins will put their courtship of...
120305  PITTSBURGH at NY GIANTS Time: 1:30 p.m. Line: ...
120306  INDIANAPOLIS -- All-Star Vince Carter was trad...
Name: review, Length: 120307, dtype: object 0          book
1          book
2          book
3          book
4          book

...
120302  sports
120303  sports
120304  sports
120305  sports
120306  sports
Name: label, Length: 120307, dtype: object

```

```
[4]: print(testing_data_topic['sentence'], testing_data_topic['topic'])
```

```

0  The atmosphere at the stadium tonight was elec...
1  The game was so intense I forgot to breathe at...
2      It had me hooked from the first chapter.
3  It's more of a slow burn than a page-turner, b...
4  It's split into two timelines, which keeps it ...
5  I could watch this film a hundred times and st...
6  Best thriller I've seen in ages. Had me on the...

```

```

7    How do you concede three goals in ten minutes?...
8    They rotated their squad for the cup game, whi...
9    The trailer gave away most of the plot, but th...
10   The protagonist was so whiny I wanted to throw...
11   The author's writing style is so unique - poet...
12   I don't get how was it supposed to work withou...
13   It's still 0-0 so far, so way too early to tel...
14   I don't get the appeal at all, it's just a cou...
15   Did you hear the screenplay for it was origina...
16   It's really incredibly impressive to mess up s...
17   The only way it's helped me is by keeping my t...

```

```
Name: sentence, dtype: object 0      sports
```

```

1    sports
2      book
3      book
4      book
5     movie
6     movie
7     sports
8     sports
9     movie
10    book
11    book
12    movie
13    sports
14    sports
15    movie
16    movie
17    book

```

```
Name: topic, dtype: object
```

```

[5]: training_data_topic['review_processed'] = training_data_topic['review'].
      ↪astype(str).map(lambda x: re.sub(r'[,\.\!?\']', '', x) if x != 'nan' else '')

# Removing 'br' from data
training_data_topic['review_processed'] =
      ↪training_data_topic['review_processed'].map(lambda x: re.sub(r'\bbr\b', '',
      ↪x))

# Convert the titles to lowercase
training_data_topic['review_processed'] =
      ↪training_data_topic['review_processed'].str.lower()

testing_data_topic['sentence'] = testing_data_topic['sentence'].str.lower()

# Print out the first rows of papers
training_data_topic['review_processed'].head()

```

```
[5]: 0    i'm only a third way in shipped lightening fas...
      1    "there have been so many times when i was scar...
      2    the media could not be loaded i personally hav...
      3    i have been a fan of britney's music since the...
      4    whether or not you're a fan it's a great read ...
      Name: review_processed, dtype: object
```

```
[6]: training_data_topic.head()
```

```
[6]:                                     review label \
0  I'm only a third way in. Shipped lightening fa...  book
1  "There have been so many times when I was scar...  book
2  The media could not be loaded. I personally ha...  book
3  I have been a fan of Britney's music since the...  book
4  Whether or not you're a fan, it's a great read...  book

                                     review_processed
0  i'm only a third way in shipped lightening fas...
1  "there have been so many times when i was scar...
2  the media could not be loaded i personally hav...
3  i have been a fan of britney's music since the...
4  whether or not you're a fan it's a great read ...
```

```
[7]: testing_data_topic.head()
```

```
[7]:                                     sentence  topic
0  the atmosphere at the stadium tonight was elec...  sports
1  the game was so intense i forgot to breathe at...  sports
2           it had me hooked from the first chapter.  book
3  it's more of a slow burn than a page-turner, b...  book
4  it's split into two timelines, which keeps it ...  book
```

0.9 Dataset statistics for topic modelling

```
[8]: import nltk
      from nltk.tokenize import word_tokenize

      nltk.download('punkt')

      def clean_text(text):
          # Remove quotation marks and empty strings
          text = re.sub(r'["`']', '', text) # Remove " and ``
          text = re.sub(r"'''", '', text) # Remove ' strings
          return text

      def truncate_text(text, head_tokens=128, tail_tokens=382, max_length=510):
          text = clean_text(text)
```

```

tokens = word_tokenize(text)
if len(tokens) > max_length:
    return " ".join(tokens[:head_tokens] + tokens[-tail_tokens:])
return text

training_data_topic['review_processed'] =
    ↪ training_data_topic['review_processed'].astype(str).map(truncate_text)
testing_data_topic['sentence'] = testing_data_topic['sentence'].astype(str).
    ↪ map(truncate_text)

train = pd.DataFrame({'text': training_data_topic['review_processed'], 'labels':
    ↪ training_data_topic['label']})
test = pd.DataFrame({'text': testing_data_topic['sentence'], 'labels':
    ↪ testing_data_topic['topic']})

# Model configuration based off of paper: Sun-etal-2020-Fine tuning BERT for
    ↪ Text Classification (pages 4 'Hyperparameters', page 5 'Dealing with long
    ↪ texts' 'Truncation methods')
# head + tail method used here

```

```

[nltk_data] Downloading package punkt to
[nltk_data]      /Users/robertostoica/nltk_data...
[nltk_data] Package punkt is already up-to-date!

```

```

[9]: # Verify data formats
print(training_data_topic[['review_processed', 'label']].head())
print(testing_data_topic[['sentence', 'topic']].head())

# Verify DataFrame formats
print(train.head())
print(test.head())
print(type(train))
print(type(test))

```

```

                                review_processed label
0  i 'm only a third way in shipped lightening fa...  book
1  there have been so many times when i was scare...  book
2  the media could not be loaded i personally hav...  book
3  i have been a fan of britney 's music since th...  book
4  whether or not you're a fan it's a great read ...  book
                                sentence  topic
0  the atmosphere at the stadium tonight was elec...  sports
1  the game was so intense i forgot to breathe at...  sports
2           it had me hooked from the first chapter.  book
3  it's more of a slow burn than a page-turner, b...  book
4  it's split into two timelines, which keeps it ...  book
                                text labels
0  i 'm only a third way in shipped lightening fa...  book

```



```

1  there have been so many times when i was scare...    book
2  the media could not be loaded i personally hav...    book
3  i have been a fan of britney 's music since th...    book
4  whether or not you're a fan it's a great read ...    book
                                     text  labels
0  the atmosphere at the stadium tonight was elec...    sports
1  the game was so intense i forgot to breathe at...    sports
2               it had me hooked from the first chapter.    book
3  it's more of a slow burn than a page-turner, b...    book
4  it's split into two timelines, which keeps it ...    book
<class 'pandas.core.frame.DataFrame'>
<class 'pandas.core.frame.DataFrame'>

```

```

[10]: print(f"Training data for BERT:\n{train.head()}")
      print(train.shape)
      print(f"Sentence example training: {train['text'][0]}")
      print('-' * 100)
      print(f"Testing data for BERT:\n{test.head()}")
      print(test.shape)
      print(f"Sentence example testing: {test['text'][0]}")

```

Training data for BERT:

```

                                     text  labels
0  i 'm only a third way in shipped lightening fa...    book
1  there have been so many times when i was scare...    book
2  the media could not be loaded i personally hav...    book
3  i have been a fan of britney 's music since th...    book
4  whether or not you're a fan it's a great read ...    book
(120307, 2)

```

Sentence example training: i 'm only a third way in shipped lightening fastthe print quality of the book is very good i like the rose and the purple foil they captured her taste here and will look bomb on my bookshelfas for the writing it 's good amazing no but they painted an incredible picture of her life much better writing than i thought to be honest i was expecting more scattered blurbs throughout the book but the way it 's presented does honor her experience and sounds like her up to an extent now i do think the ghost writer went verbose in some areas that were unnecessary which derailed her voice in my head but i imagine getting the whole picture would be difficulti was hoping for a thicker on her for having convictioni suddenly found myself understanding little quirks connecting the dots certain things would suddenly click in my head for example her british accent and her relationship with a lynchpin matriarch eldervery cool build up laying the ground work for her future decisions by presenting us with complex childhood family dynamics really has me understanding her choices and quirks much better than i ever realized i needed to be deep in thought ofsometimes we judge from a place of pity or outsider perspective instead of compassion now i see the soft gentle feminine side of her mindset that we knew when she was younger start to peek through the pages it 's that sweet part of her soul that she does n't show much nowadays for good reason of course as she

's been deeply hurt and damaged by other 's greed but that delicate nature is part of why most of us were charmed by her i hope she finds her way back one day she 's been through a lot and i found myself relating to things she 'd been through and feeling deeply saddened by her sense of insecurity in childhood her insights correlate along with many of my own life experiences and traumas i was taken aback by her growth in reflection as one would believe anger would be the easy way to portray her story not the case here it 's her appreciation for her past that has earned my deep respect beyond what i thought i had possessed prior to reading her story with her brother was touching and softened my distain for his participation in the matter of handling her affairs although i hope he is held accountable someday it shows how loving and innocent she was and how grave a loss she has suffered for it is those who are selfless that do n't speak up only to suffer the most such a contrast to the presentation of her character in the media thus far this book is an excellent read -- -- -- to be continued * i 'm looking forward to reading through this book in it 's entirety and will update when i 'm done one star for the writing not sounding like her in some places i will change accordingly

Testing data for BERT:

	text	labels
0	the atmosphere at the stadium tonight was elec...	sports
1	the game was so intense i forgot to breathe at...	sports
2	it had me hooked from the first chapter.	book
3	it's more of a slow burn than a page-turner, b...	book
4	it's split into two timelines, which keeps it ...	book

(18, 2)

Sentence example testing: the atmosphere at the stadium tonight was electric.

```
[11]: train_label_counts = Counter(train['labels'])

test_label_counts = Counter(test['labels'])

print("Class distribution in training data:", train_label_counts)
print("Class distribution in testing data:", test_label_counts)

#Class distribution in training data: Counter({'movie': 50000, 'sports': 36726,
↪ 'book': 33581})
#Class distribution in testing data: Counter({'sports': 6, 'book': 6, 'movie': 6})
↪ 6})
```

Class distribution in training data: Counter({'movie': 50000, 'sports': 36726, 'book': 33581})

Class distribution in testing data: Counter({'sports': 6, 'book': 6, 'movie': 6})

```
[12]: long_string = ','.join(list(training_data_topic['review_processed'].values))
```

[12] :



[13] :

```

plt.ylabel('counts')
plt.show()

# Initialise the count vectorizer with the English stop words
count_vectorizer = CountVectorizer(stop_words='english')

# Fit and transform the processed titles
count_data = count_vectorizer.
    ↪fit_transform(training_data_topic['review_processed'])

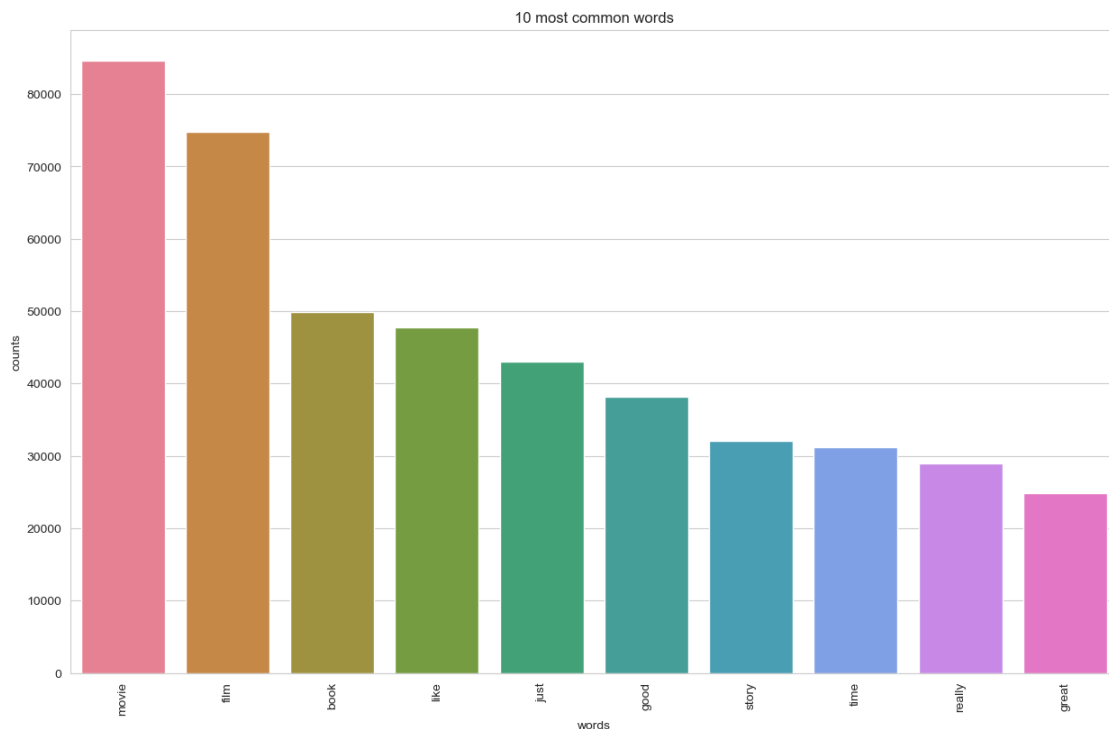
# Visualise the 10 most common words
plot_10_most_common_words(count_data, count_vectorizer)

```

/var/folders/lc/kpr2rnx93q52qglqhvd6c6vw0000gn/T/ipykernel_79064/2938581420.py:2
0: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

```
sns.barplot(x=x_pos, y=counts, palette='husl')
```



0.10 Model for topic classification

```
[14]: max_samples = 10000

train = train.groupby('labels', group_keys=False).apply(lambda x: x.
    ↪sample(n=min(len(x), max_samples), random_state=42))

print(train['labels'].value_counts())
```

```
labels
book      10000
movie     10000
sports    10000
Name: count, dtype: int64
```

```
/var/folders/lc/kpr2rn93q52qglqhvd6c6vw0000gn/T/ipykernel_79064/3045598224.py:3
: DeprecationWarning: DataFrameGroupBy.apply operated on the grouping columns.
This behavior is deprecated, and in a future version of pandas the grouping
columns will be excluded from the operation. Either pass `include_groups=False`
to exclude the groupings or explicitly select the grouping columns after groupby
to silence this warning.
```

```
train = train.groupby('labels', group_keys=False).apply(lambda x:
x.sample(n=min(len(x), max_samples), random_state=42))
```

```
[15]: from sklearn.model_selection import train_test_split

train, dev = train_test_split(train, test_size=0.1, random_state=0,
                             stratify=train[['labels']])
```

```
[16]: train['labels'] = train['labels'].astype('category')
dev['labels'] = dev['labels'].astype('category')
train['text'] = train['text'].str.strip()
dev['text'] = dev['text'].str.strip()

train['labels'] = train['labels'].str.strip()
dev['labels'] = dev['labels'].str.strip()
```

```
[17]: from sklearn.preprocessing import LabelEncoder

# Convert labels to integer format
label_encoder = LabelEncoder()

train['labels'] = label_encoder.fit_transform(train['labels'])
dev['labels'] = label_encoder.transform(dev['labels'])

print(train.head())
```

```
                                text  labels
104792  baku : david beckham is set to be hauled over ...          2
```

12766		go slu	0
9880	a more accurate title for this book would be "...		0
77921	i saw two hands back in sydney a few years ago...		1
103179	the first two at-bats were the kind of moments...		2

```
[18]: # Flatten any potential multi-dimensional array (if present)
train['labels'] = train['labels'].apply(lambda x: x[0] if isinstance(x, list)
    ↪ else x)
dev['labels'] = dev['labels'].apply(lambda x: x[0] if isinstance(x, list) else
    ↪ x)

print(train.head())
```

		text	labels
104792	baku : david beckham is set to be hauled over ...		2
12766		go slu	0
9880	a more accurate title for this book would be "...		0
77921	i saw two hands back in sydney a few years ago...		1
103179	the first two at-bats were the kind of moments...		2

```
[19]: print(len(train))
print("train:", train[['labels']].value_counts(sort=False))
print(train.head(3))
print(type(train))
```

```
27000
train: labels
0      9000
1      9000
2      9000
Name: count, dtype: int64
```

		text	labels
104792	baku : david beckham is set to be hauled over ...		2
12766		go slu	0
9880	a more accurate title for this book would be "...		0

```
<class 'pandas.core.frame.DataFrame'>
```

```
[20]: print(len(dev))
print("dev:", dev[['labels']].value_counts(sort=False))
print(dev.head(3))
print(type(dev))
```

```
3000
dev: labels
0      1000
1      1000
2      1000
Name: count, dtype: int64
```

		text	labels
--	--	------	--------

```

29094    i'm sorry- no offense meant to the writer- but...      0
45019    nick millard aka nick phillips should have lef...    1
106052    ap - rich gannon will not return to the field ...    2
<class 'pandas.core.frame.DataFrame'>

```

```

[26]: # Model configuration
bert_args = ClassificationArgs()

bert_args.overwrite_output_dir=True
bert_args.evaluate_during_training=True

bert_args.num_train_epochs=10
bert_args.train_batch_size=32
bert_args.gradient_accumulation_steps = 4
bert_args.learning_rate=5e-5
bert_args.max_seq_length=256

bert_args.warmup_ratio=0.1

bert_args.use_early_stopping=True
bert_args.early_stopping_delta=0.01

bert_args.early_stopping_metric='eval_loss'
bert_args.early_stopping_metric_minimize=True
bert_args.early_stopping_patience=8
bert_args.evaluate_during_training_steps=32

# Model configuration based off of paper: Sun-etal-2020-Fine tuning BERT for
↳Text Classification (pages 4 'Hyperparameters', page 5 'Catastrophic
↳Forgetting')
# and due to the needs of the training data labeled as big_data.csv

```

```

[27]: steps_per_epoch = int(np.ceil(len(train) / float(bert_args.train_batch_size)))
print('Each epoch will have {:,} steps.'.format(steps_per_epoch))

```

Each epoch will have 844 steps.

```

[28]: model = ClassificationModel('bert', 'bert-base-cased', num_labels=3,
↳args=bert_args, use_cuda=False)

```

Some weights of BertForSequenceClassification were not initialized from the model checkpoint at bert-base-cased and are newly initialized:

```
['classifier.bias', 'classifier.weight']
```

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

```

[29]: print(str(model.args).replace(',', '\n'))

```

```
ClassificationArgs(adafactor_beta1=None
```

```

adafactor_clip_threshold=1.0
adafactor_decay_rate=-0.8
adafactor_eps=(1e-30
0.001)
adafactor_relative_step=True
adafactor_scale_parameter=True
adafactor_warmup_init=True
adam_betas=(0.9
0.999)
adam_epsilon=1e-08
best_model_dir='outputs/best_model'
cache_dir='cache_dir/'
config={}
cosine_schedule_num_cycles=0.5
custom_layer_parameters=[]
custom_parameter_groups=[]
dataloader_num_workers=0
do_lower_case=False
dynamic_quantize=False
early_stopping_consider_epochs=False
early_stopping_delta=0.01
early_stopping_metric='eval_loss'
early_stopping_metric_minimize=True
early_stopping_patience=8
encoding=None
eval_batch_size=100
evaluate_during_training=True
evaluate_during_training_silent=True
evaluate_during_training_steps=32
evaluate_during_training_verbose=False
evaluate_each_epoch=True
fp16=False
gradient_accumulation_steps=4
learning_rate=5e-05
local_rank=-1
logging_steps=50
loss_type=None
loss_args={}
manual_seed=None
max_grad_norm=1.0
max_seq_length=256
model_name='bert-base-cased'
model_type='bert'
multiprocessing_chunksize=-1
n_gpu=1
no_cache=False
no_save=False
not_saved_args=[]

```



```

num_train_epochs=10
optimizer='AdamW'
output_dir='outputs/'
overwrite_output_dir=True
polynomial_decay_schedule_lr_end=1e-07
polynomial_decay_schedule_power=1.0
process_count=10
quantized_model=False
reprocess_input_data=True
save_best_model=True
save_eval_checkpoints=True
save_model_every_epoch=True
save_optimizer_and_scheduler=True
save_steps=2000
scheduler='linear_schedule_with_warmup'
silent=False
skip_special_tokens=True
tensorboard_dir=None
thread_count=None
tokenizer_name='bert-base-cased'
tokenizer_type=None
train_batch_size=32
train_custom_parameters_only=False
trust_remote_code=False
use_cached_eval_features=False
use_early_stopping=True
use_hf_datasets=False
use_multiprocessing=True
use_multiprocessing_for_evaluation=True
wandb_kwargs={}
wandb_project=None
warmup_ratio=0.1
warmup_steps=0
weight_decay=0.0
model_class='ClassificationModel'
labels_list=[0
1
2]
labels_map={}
lazy_delimiter='\t'
lazy_labels_column=1
lazy_loading=False
lazy_loading_start_line=1
lazy_text_a_column=None
lazy_text_b_column=None
lazy_text_column=0
onnx=False
regression=False

```

```

sliding_window=False
special_tokens_list=[]
stride=0.8
tie_value=1)

```

```
[30]: _, history = model.train_model(train, eval_df=dev)
```

```

0%|          | 0/54 [00:00<?, ?it/s]

```

huggingface/tokenizers: The current process just got forked, after parallelism has already been used. Disabling parallelism to avoid deadlocks...

To disable this warning, you can either:

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false)
Epoch:  0%|          | 0/10 [00:00<?, ?it/s]
Running Epoch 1 of 10:  0%|          | 0/844 [00:00<?, ?it/s]
0%|          | 0/6 [00:00<?, ?it/s]
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- Explicitly set the environment variable      - Avoid using
`tokenizers` before the fork if possible

```

```

- Explicitly set the environment variable huggingface/tokenizers: The
current process just got forked, after parallelism has already been used.
Disabling parallelism to avoid deadlocks...
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TOKENIZERS_PARALLELISM - Avoid using `tokenizers` before the fork if possible
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TOKENIZERS_PARALLELISM
TOKENIZERS_PARALLELISM=(true | false)
=(true | false)
To disable this warning, you can either:
- Avoid using `tokenizers` before the fork if possible
- Explicitly set the environment variable TOKENIZERS_PARALLELISM=(true |
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0%|          | 0/6 [00:00<?, ?it/s]

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0%|          | 0/6 [00:00<?, ?it/s]

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```

```

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    - Explicitly set the environment variable TOKENIZERS_PARALLELISM=(true |
false)

```



```

Running Epoch 2 of 10:   0%|               | 0/844 [00:00<?, ?it/s]

   0%|               | 0/6 [00:00<?, ?it/s]

huggingface/tokenizers: The current process just got forked, after parallelism
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   0%|               | 0/6 [00:00<?, ?it/s]

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```

```

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```

```

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0%|          | 0/6 [00:00<?, ?it/s]

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TOKENIZERS_PARALLELISMhuggingface/tokenizers: The current process just got
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To disable this warning, you can either:
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```

```
0%|          | 0/6 [00:00<?, ?it/s]
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Running Epoch 3 of 10:  0%|          | 0/844 [00:00<?, ?it/s]

  0%|          | 0/6 [00:00<?, ?it/s]

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```

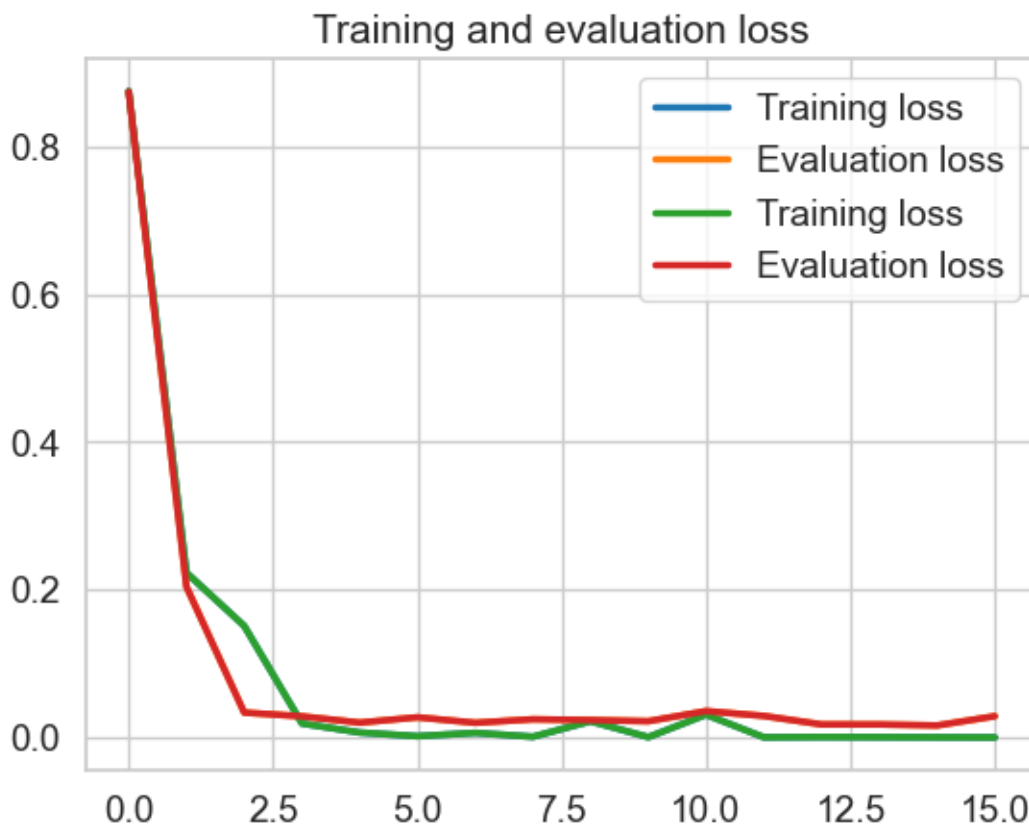
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```
[46]: train_loss = history['train_loss']
eval_loss = history['eval_loss']
plt.plot(train_loss, label='Training loss')
plt.plot(eval_loss, label='Evaluation loss')
plt.title('Training and evaluation loss')
plt.legend()
plt.show()
```



```
[32]: result, model_outputs, wrong_predictions = model.eval_model(dev)
result
```



```
0%|          | 0/6 [00:00<?, ?it/s]
```

```
huggingface/tokenizers: The current process just got forked, after parallelism
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```
Running Evaluation: 0%|          | 0/30 [00:00<?, ?it/s]
```

```
[32]: {'mcc': np.float64(0.9895051124562882), 'eval_loss': 0.02902314242965076}
```

```
[33]: predicted, probabilities = model.predict(test.text.to_list())
      test['predicted'] = predicted
```

```
0it [00:00, ?it/s]
```

```
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```
0%|          | 0/1 [00:00<?, ?it/s]
```

```
[56]: test['predicted'] = label_encoder.inverse_transform(test['predicted'])
```

```
-----
ValueError                                Traceback (most recent call last)
Cell In[56], line 1
----> 1 test['predicted'] = label_encoder.inverse_transform(test['predicted'])

File ~/Desktop/anaconda3/envs/Text_Mining_python/lib/python3.12/site-packages/
sklearn/preprocessing/_label.py:162, in LabelEncoder.inverse_transform(self, y)
    156 diff = _setdiff1d(
    157     ar1=y,
    158     ar2=xp.arange(self.classes_.shape[0], device=device(y)),
    159     xp=xp,
    160 )
    161 if diff.shape[0]:
--> 162     raise ValueError("y contains previously unseen labels: %s" %
    ↪str(diff))
    163 y = xp.asarray(y)
    164 return xp.take(self.classes_, y, axis=0)

ValueError: y contains previously unseen labels: ['book' 'movie' 'sports']
```

```
[57]: print(test.head(5))
```

	text	labels	predicted
0	the atmosphere at the stadium tonight was elec...	sports	sports
1	the game was so intense i forgot to breathe at...	sports	book
2	it had me hooked from the first chapter.	book	book
3	it's more of a slow burn than a page-turner, b...	book	book
4	it's split into two timelines, which keeps it ...	book	book

```
[58]: print(train['labels'].dtype)
print(train['labels'].unique())
```

```
object
['sports' 'book' 'movie']
```

```
[61]: print(classification_report(test['labels'], test['predicted']))
```

	precision	recall	f1-score	support
book	0.45	0.83	0.59	6

movie	1.00	0.17	0.29	6
sports	0.83	0.83	0.83	6
accuracy			0.61	18
macro avg	0.76	0.61	0.57	18
weighted avg	0.76	0.61	0.57	18